Supervised Contrastive Learning Based on Fusion of Global and Local Features for Remote Sensing Image Retrieval

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***Abstract*— With the rapid development of remote sensing sensor technology, the number of remote sensing images (RSIs) has exploded. How to effectively retrieve and manage this massive data have become an urgent problem. At present, content-based image retrieval (CBIR) methods have become a mainstream method due to their excellent performance. However, most of the existing retrieval methods only consider the global features of images, which lacks the ability to discriminate images with the same semantic information but different visual representations. To alleviate this issue, supervised contrastive learning based on the fusion of global and local features method is proposed in this article, named SCFR. First, a fusion module is designed to combine global and local features to enhance the ability of image expression. Second, supervised contrastive learning is introduced into the retrieval task to effectively improve the feature distribution, so that the positive sample pairs are close to each other, and the negative sample pairs are far away from each other in the feature space. Furthermore, to make the distribution of features of the same class more compact, the center contrastive loss is added to the constraints and combines the class centers that change iteratively with the network. Experimental results on three RSI datasets show that our proposed method has a more effective retrieval performance than the state-of-the-art methods. The code and models are available at https://github.com/xdplay17/SCFR.**

***Index Terms*— Contrastive learning, hash learning, image retrieval, remote sensing.**

I. Introduction

With the comprehensive development of remote sensing sensor technology, the number of remote sensing images (RSIs) has shown explosive growth, which promotes multiple applications such as environmental detection, geological survey, and mapping [1]. However, when it comes to a specific application, a formidable challenge is to efficiently and effectively search for images matching the query content in a large collection of RSIs. Recently, RSI retrieval (RSIR) methods have received extensive attention as an important research orientation for searching for nearest neighbor information.

Early RSIR methods manually labeled images with texts, thus similar images can be found by text annotations. Unfortunately, heavy use of manpower and inaccurate retrieval results contribute to the demise of text-based image retrieval methods. In recent years, RSIR methods, which contain two pivotal steps of feature extraction and similarity measurement, have been developed rapidly because of their advantage of automatically and quickly extracting core information from images. Among them, feature extraction principally generates three levels of features, including low-level features, middle-level features, and high-level features. In detail, texture, spectrum, and shape [2], [3], [4] are the main information contained in low-level features. The middle-level features are generally obtained by combining low-level features in some coding ways, including a bag of words [5], fisher vector [6], and vector vocally aggregated descriptors [7]. However, low-level and middle-level features designed by handcraft have drawbacks which are poor representation ability, resulting in limited performance in content-based image retrieval (CBIR).

Owing to the development of deep learning technologies, deep neural networks such as AlexNet [8], visual geometry group (VGG) [9], and ResNet [10] extract high-level features serving as a signature of image semantic content. High-level features have gradually become the key descriptors used in CBIR tasks due to their powerful expressive ability. Cao et al. [11] employ a feature extraction structure constructed by convolutional and fully connected layers to acquire high-level features. Liu et al. [12] devise a deep learning model with feature aggregation and attention branch, which can capture multiscale and discriminative information. However, most CBIR methods focus on representing RSIs with global features but ignore the attention to local features. In fact, local features contain discriminative geometric information about the image region, which cannot be provided by global features. Yang et al. [13] propose an orthogonal fusion framework to aggregate global and local features. Peng et al. [14] fuse local and global representations under different resolutions via the module of the feature coupling unit. Both of them achieve information complementarity and acquire high-quality features. However, there is still room for improvement by

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combining global features and local features to enhance the ability to express images.

In CBIR tasks, in addition to designing networks to obtain efficient expression features, similarity measurement is also very critical, which greatly affects retrieval performance. The literature [12], [15] both design a pairwise similarity loss function to ensure that the similarities between the RSIs in the original feature space can be retained in the hashing space. Song et al. [16] adopt the product loss between the query images and the database images to learn the similarity-preserving deep features. The losses all optimize the distribution of features and enhance the reliability of similarity measurement. Unfortunately, these loss functions commonly used in RSIR tasks cannot achieve the optimal retrieval effect.

Once representative and distinguishing features are obtained, the problems of large storage space requirements and high computational resource consumption have to be solved because the dimension of the feature is high. Therefore, the hash learning method is proposed to map high-dimensional feature vectors to low-dimensional feature vectors. The low-dimensional feature vector is binarized into a hash code to participate in storage and calculation, which greatly alleviate the abovementioned problems. At present, deep hashing has become the mainstream solution for RSIR tasks. Song et al. [17] use a hash layer perfectly inserted into the network to transfer the deep features into compact hash codes. Li et al. [18] develop quantized deep learning to hash framework for large-scale RSIR. It is worth noting that due to the particularity of RSIs, some images with the same semantic features may have great visual differences. The existing retrieval methods often make these images with the same semantic features in the feature space far away from each other and misclassified into different classes. And the accuracy of similarity measurement is reduced, which leads to incorrect retrieval results.

To alleviate the above issues, we propose a deep hashing convolutional neural network framework based on supervised contrastive learning to improve the discrimination among image features (named SCFR), which integrates global and local features to fully express the key information of images. First, a global and local feature fusion module is designed to improve the image expression ability. In the local feature extraction module, a spatial attention mechanism is used to provide discriminative information that cannot be characterized by global features. At the same time, the global feature module extracts semantic information from images. Second, supervised contrastive learning is introduced in this article, which uses label information to construct positive and negative samples for each sample. Then, the loss function is constructed so that the positive samples of similar samples are close and the negative samples are far away from each other. The advantage is that the spatial distribution of RSI features can be optimized. Finally, to further improve the performance of RSI hash codes, feature centers cooperate with the center contrastive loss, which are updated with the iteration to make the features of the same class more compact. The main contributions involved in this article can be summarized as follows.

1) To improve the ability of image representation, a module combining global and local features is designed.
2) Supervised contrastive learning is introduced for similarity measurement. In the embedding space, features belonging to the same class are pulled together, and features from different classes are separated.
3) The center contrastive loss is used to calculate the feature centers, which has the benefit of making similar features more compact to generate high-quality hash codes.
4) Experiments on three datasets show that the proposed SCFR can outperform other methods to achieve state-of-the-art performance in RSI retrieval.

The remainder of this article is described as follows. Section II briefly introduces the related work. Section III shows the different parts of the proposed method in detail. Comprehensive experiments and discussions are exhibited in Section IV. Finally, conclusions are presented in Section V.

II. Related Works

A. Contrastive Learning

In recent years, contrastive learning has caused a boom in many tasks attributed to its capacity for intraclass compactness and interclass separability. The main idea of contrastive learning is to construct positive and negative samples of an anchor through data augmentation methods such as Cutmix [19] and Mixup [20], and then the anchor is close to its positive samples and far away from its negative samples in the feature space by optimizing the contrastive loss function. Contrastive learning can usually be divided into unsupervised contrastive learning and supervised contrastive learning. Unsupervised contrastive learning maximizing the instance discriminativeness has shown its high performance across various downstream tasks. For example, Wu et al. [21] propose an unsupervised feature learning approach, which maximizes distinction between instances via a novel contrastive loss. Li et al. [22] utilize unsupervised contrastive learning to acquire effective facial expression features in the facial expression recognition task. However, unsupervised contrastive learning blindly stretches the distance between two instances in feature space even if they are from the same class.

Compared with unsupervised learning, supervised contrastive learning has prior information on labels. Labels cannot only help deep neural network model training to obtain features with semantic information but also help us construct positive and negative samples of anchors. Specifically, samples with the same label as the anchor are positive samples, and samples with different labels are negative samples. Under the condition of utilizing label information, contrastive learning makes the features of the same class close to each other and the features of different classes away. For example, Cui et al. [23] propose parametric contrastive learning to push samples of the same class close with a set of parametric learnable centers. Zhu et al. [24] propose balanced contrastive learning to achieve intraclass compactness and interclass separability for long-tailed data.

In addition to studying the methods of contrastive learning, how to correctly understand contrastive learning has

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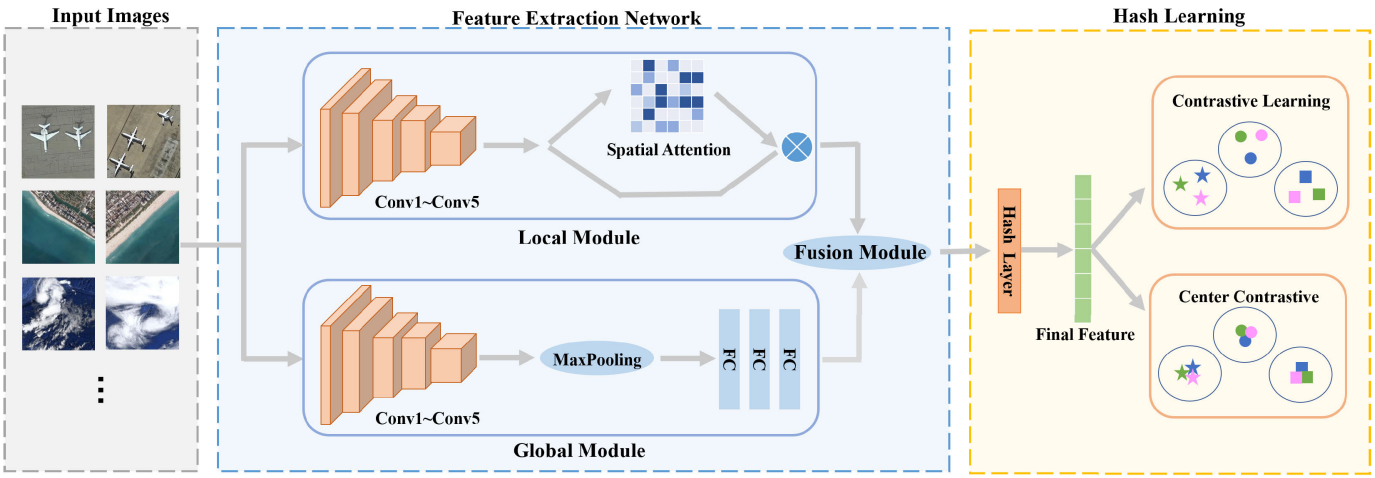


Fig. 1. Framework of our method SCFR.

also attracted increasing attention. Prakash and Gupta [25] demystify contrastive self-supervised learning by presenting a framework to evaluate invariances in representations. Chen et al. [26] find three interesting properties of contrastive learning. First, they show that there is little difference between contrastive learning losses with a multilayer projection head. Second, contrastive learning can learn meaningful local features from instances despite operating on global image representations. Third, simple features are easier to learn, and they prevent the network from learning other more complex features. Wang and Liu [27] reveal that contrastive loss is a hardness-aware loss, and the temperature parameter of the loss function plays a significant role in controlling the strength of penalties on hard negative samples.

B. Hash Learning

For fast and accurate indexing and searching, hash learning is widely used in large-scale image retrieval tasks. Hash learning attempts to map data into a low-dimensional hash space, which saves a lot of memory space and computational resources. Preserving the semantic similarity of the raw data in the hash space is the key to the success of hash learning. Existing hash learning methods can be mainly divided into two subgroups: unsupervised hash learning and supervised hash learning. Unsupervised hash learning maps unlabeled data to compact binary codes. For instance, spectral hashing [28] calculates the hashing code by thresholding a subset of Laplacian eigenvectors for similarity graphs. Iterative quantization [29] proposes to learn binary codes by finding the rotation of zero-centered data, which uses an alternating minimization algorithm to minimize the quantization error of mapping input data to codewords. Density-sensitive hashing [30] tries to rely on the geometry of the data for more discriminative hash codes. However, unsupervised hash learning performs poorly for large and complex tasks without the help of prior information.

On the contrary, supervised hash learning can obtain high-quality hash codes by relying on the hash map function learned from supervised information. Sparse embedding and least variance encoding [31] obtains high-performance binary codes by employing a sparse embedding and least variance encoding method. Minimal loss hashing [32] develops a new loss to train structural SVMs for similarity-preserving binary codes. Binary reconstructive embeddings [33] constructs hash functions by minimizing a squared loss generated by the difference between the original distances and the Hamming distances. Supervised hashing with kernels [34] proposes a kernel-based supervised hashing method in which the distance of similar pairs is minimized and the distance of dissimilar pairs is maximized in the Hamming space. However, all of these hashing algorithms use handcrafted visual features to represent the images, which reduces the performance of the hash algorithm because handcrafted visual features have a limited ability of expression.

In recent years, with the development of deep learning, supervised hash learning is used more with neural networks. For example, a deep hashing network (DHN) [35] designs a pairwise cross-entropy loss and a quantization loss to optimize deep neural networks which improves the quality of hash coding. Deep Cauchy hashing (DCH) [36] devises a pairwise cross-entropy loss function inspired by the Cauchy distribution, which reduces Hamming distance of similar image pairs. Central similarity quantization (CSQ) [37] predefines a set of separate hash centers, optimizing the features of data points to be close to their corresponding hash centers.

III. Proposed Method

The retrieval framework designed in this article is shown in Fig. 1. First, a two-branch feature fusion network is proposed to extract the features of images, which includes a global branch module, a local branch module, and a feature fusion module. The global branch and the local branch extract global features and local features, respectively, and the feature fusion module fuses these two kinds of features into compact features. Second, contrastive learning is introduced to achieve intraclass compactness and interclass dissimilarity, which constructs positive and negative samples through label information to compare features and features, as well as feature and feature

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centers. Third, the compact features are binarized into hash codes, and the Hamming distance is used to measure the similarity of RSIs.

A. SCFR Framework

Considering the limitation of information contained in global features, local features are extracted and fused with global features to achieve complementarity of feature information.

1) Global Module: In Fig. 1, AlexNet is chosen as the backbone network. Similar to [16], the classification layer of AlexNet is discarded, and the remaining structure is used to extract the global features containing the semantic information of RSIs. Specifically, suppose there are N training images, $X = \{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_N\}$ and $Y = \{\mathbf{y}_1, \mathbf{y}_2, \ldots, \mathbf{y}_N\}$ indicate the labels of the corresponding images, where $i \in Q = \{1, 2, \ldots, N\}$ are the indexes of the images. Then, all training images are passed through five convolutional layers (conv1–conv5) and two fully connected layers (fc6 and fc7) to obtain a set of high-dimensional depth global features $\mathbf{f}_g = \{\mathbf{f}_{g,1}, \mathbf{f}_{g,2}, \ldots, \mathbf{f}_{g,N}\}$. For any $\mathbf{x}_i$, its global feature vector $\mathbf{f}_{g,i} \in \mathbb{R}^{1\times 256}$ can be denoted as

$$\mathbf{f}_{g,i} = \Phi(\mathbf{x}_i; \mathbf{W}_g), \quad i = 1, 2, \ldots, N \tag{1}$$

where $\mathbf{W}_g$ is the parameter of the first seven-layer structure of AlexNet. And Φ is a mapping function that is characterized by $\mathbf{W}_g$, including some operations such as convolution and pooling.

2) Local Module: The local features $\mathbf{f}_l = \{\mathbf{f}_{l,1}, \mathbf{f}_{l,2}, \ldots, \mathbf{f}_{l,N}\}$ can be extracted from the conv5 layer of AlexNet. To fully learn the spatial distribution information of the image, the proposed method in Fig. 1 introduces an attention module into the local feature extraction branch. Specifically, the local feature map extracts the background and texture information of the image through the maximum and the average pooling operation, respectively. The resulting features are then fused through a 1×1 convolutional layer. Finally, the spatial attention map is obtained through the sigmoid operation. Local feature $\mathbf{f}_l \in \mathbb{R}^{H\times W\times 256}$ obtained by the spatial attention mechanism can be expressed as

$$\mathbf{f}_{l,i} = \Psi(\mathbf{x}_i; \mathbf{W}_l), \quad i = 1, 2, \ldots, N \tag{2}$$

where $\mathbf{W}_l$ is composed of the parameters of the local module. And Ψ obtains the attention map about $\mathbf{x}_i$ through operations such as convolution, pooling, and activation.

3) Fusion Module: In the two branches described above, the global feature mainly contains the semantic information of the image, while the local feature mainly contains the discriminative geometric information of a specific image area. Hence, a fusion module is naturally designed to learn more representative features. First, the global feature $\mathbf{f}_g$ is resized to the size of the local feature $\mathbf{f}_l$ to solve the mismatch between the dimension of $\mathbf{f}_g \in \mathbb{R}^{1\times 256}$ and $\mathbf{f}_l \in \mathbb{R}^{H\times W\times 256}$. Second, the expanded global feature and corresponding local feature are concatenated together, and then, a pooling layer is used to complete feature fusion. Mathematically, the fusion process can be formulated as

$$\mathbf{f}_{\text{fuse},i} = \text{pool}(\text{cat}(\text{expand}(\mathbf{f}_{g,i}); \mathbf{f}_{l,i})), \quad i = 1, 2, \ldots, N \tag{3}$$

where expand, cat, and pool operations, respectively, represent the expansion of data dimensions, data concatenation, and data pooling.

4) Hash Learning Module: A hash layer with K nodes is added at the end of the network to get the hash codes $B = \{\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_N\} \in \{+1, -1\}^{N\times K}$ of the RSIs for more efficient retrieval, where K determines the number of bits in the hash code. Due to the nonderivability of the sign$(\cdot)$ function, tanh$(\cdot)$ function replaces it as the activation function after the hash layer during the training process. Therefore, the real-valued hash codes $H = \{\mathbf{h}_1, \mathbf{h}_2, \ldots, \mathbf{h}_N\}$ during training can be expressed as

$$\mathbf{h}_i = \tanh(\mathbf{W}_h \mathbf{f}_{\text{fuse},i} + \mathbf{v}_h), \quad i = 1, 2, \ldots, N \tag{4}$$

where $\mathbf{W}_h \in \mathbb{R}^{K\times 512}$ and $\mathbf{v}_h \in \mathbb{R}^{K\times 1}$ are the weights and biases of the hash layer, respectively. Since the inner product is employed to measure the cosine similarity of features in contrastive learning, it is necessary to normalize the real-valued hash codes $H = \{h_1, h_2, \ldots, h_N\}$ to the hypersphere to obtain the normalized feature vectors $Z = \{z_1, z_2, \ldots, z_N\}$. For each normalized feature vector, each element of it can be expressed as

$$z_{ik} = \frac{h_{ik}}{\|\mathbf{h}_i\|_2}, \quad k = 1, 2, \ldots, K. \tag{5}$$

After training, the hash code is obtained by binarizing the standardized feature $\mathbf{z}_i$, which can be expressed as

$$\mathbf{b}_i = \text{sign}(\mathbf{z}_i), \quad i = 1, 2, \ldots, N. \tag{6}$$

B. Loss Function Definition

In real RSIs, some images with the same semantic features may have large differences in vision, resulting in a long distance in the feature space, and incorrect retrieval results. To alleviate this problem, the supervised contrastive learning method is used to increase intraclass compactness and increase interclass dissimilarity, which can further enhance the discriminant ability of features to improve retrieval performance. According to [38], the feature representation loss of contrastive learning can be expressed as

$$\mathcal{L} = -\sum_{i\in P} \log \frac{\exp\left(\mathbf{b}_i \cdot \mathbf{b}_{j(i)}^T / \tau\right)}{\sum_{r\in R(i)} \exp\left(\mathbf{b}_i \cdot \mathbf{b}_r^T / \tau\right)} \tag{7}$$

where $P = \{1, 2, \ldots, 2N\}$ represents the index of a set of arbitrary augmented RSIs and $R(i) = P\backslash\{i\}$. In unsupervised learning, the positive sample pairs are constructed by some data augmentation methods. For instance, $\mathbf{b}_i$ and $\mathbf{b}_{j(i)}$ are positive sample pairs generated from the same RSI using two different data augmentation strategies. When $\mathbf{b}_i$ is an anchor, $\mathbf{b}_{j(i)}$ is a positive sample belonging to $\mathbf{b}_i$, and the other $2N - 2$ samples are negative samples of $\mathbf{b}_i$.

However, label information can be leveraged to construct positive and negative samples for contrastive learning under supervised conditions. Due to the addition of label information, the index of the image with the same label as $\mathbf{b}_i$ is used to establish the index set $S(i)$ of the positive samples of $\mathbf{b}_i$.

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Conversely, the images with a different label than $\mathbf{b}_i$ are its negative samples. Thus, (7) can be rewritten as

$$\mathcal{L}_{\text{con}} = \sum_{i\in P} \frac{-1}{n(S(i))} \sum_{s\in S(i)} \log \frac{\exp\left(\mathbf{b}_i \cdot \mathbf{b}_s^T/\tau\right)}{\sum_{r\in R(i)} \exp\left(\mathbf{b}_i \cdot \mathbf{b}_r^T/\tau\right)} \tag{8}$$

where $n(S(i))$ represents the number of indexes contained in $S(i)$. It is noteworthy that images cannot be retrieved after data augmentation in the image retrieval task. At this time, if the augmented images are used for training, the retrieval performance will be reduced. Therefore, it is necessary to remove the augmented data strategy during training. Equation (8) can be updated as

$$\mathcal{L}_{\text{con}} = \sum_{i\in Q} \frac{-1}{n(S(i)) + \varepsilon} \sum_{s\in S(i)} \log \frac{\exp\left(\mathbf{b}_i \cdot \mathbf{b}_s^T/\tau\right)}{\sum_{r\in R(i)} \exp\left(\mathbf{b}_i \cdot \mathbf{b}_r^T/\tau\right)} \tag{9}$$

in which ε is set to a very small positive value to prevent the case where $n(S(i))$ is 0.

In (9), it is a discrete problem to directly learn the feature distribution of the hash code, which often leads to the failure of normal backpropagation of the gradient when training the network. To avoid the optimization discrete problem, the hash code $\mathbf{b}$ is replaced by the feature $\mathbf{z}$

$$\mathcal{L}_{\text{con}} = \sum_{i\in Q} \frac{-1}{n(S(i)) + \varepsilon} \sum_{s\in S(i)} \log \frac{\exp\left(\mathbf{z}_i \cdot \mathbf{z}_s^T/\tau\right)}{\sum_{r\in R(i)} \exp\left(\mathbf{z}_i \cdot \mathbf{z}_r^T/\tau\right)}. \tag{10}$$

The network trained with the contrastive learning loss has been able to achieve good intraclass compactness and interclass difference in the feature space. Therefore, it is feasible to learn a center p_c for each class c, which can make features belonging to this class close to each other and features not belonging to this class far away. Based on the above analysis, a center-gathering strategy is introduced to further optimize intraclass compactness and interclass dissimilarity. First, the mean of feature vectors corresponding to each class is used to create the set of class centers $P = \{\mathbf{p}_1, \ldots, \mathbf{p}_C\}$, which can be calculated as

$$\mathbf{p}_c = \frac{\sum_{m=1}^{M} \mathbf{z}_i (\mathbf{z}_i \in c)}{M} \tag{11}$$

where M represents the number of images each class has in a set of samples. On the basis of (11), an additional central constraint is added to learn a better feature space distribution, i.e.,

$$\mathcal{L}_{\text{cen}} = \sum_{i=1}^{N} \sum_{c=1}^{C} l(\mathbf{z}_i, \mathbf{p}_c) \tag{12}$$

$$l(\mathbf{z}_i, \mathbf{p}_c) = \begin{cases} D(\mathbf{z}_i, \mathbf{p}_c), & \text{class}(\mathbf{z}_i) = c \\ \max\{0, \Delta - D(\mathbf{z}_i, \mathbf{p}_c)\}, & \text{otherwise} \end{cases} \tag{13}$$

in which Δ is a margin hyperparameter that defines how close a similar and dissimilar item can be, and D represents Hamming distance function.

Since the use of the sign$(\cdot)$ function to directly convert the real-valued hash codes into the binary hash codes causes a non-negligible performance loss, a hash quantization function is designed to perform binary quantization for real-valued hash codes H. The bimodal Laplacian prior is introduced to push each element in h_i approach 1 or -1. The quantization loss function of the proposed method SCFR can be expressed as

$$\mathcal{L}_{\text{qua}} = \frac{\sum_{i=1}^{N} \sum_{k=1}^{K} (|h_{ik}| - 1)^2}{N}. \tag{14}$$

Finally, combining (10), (12), and (14), the overall optimization loss function can be written as

$$\mathcal{L} = \mathcal{L}_{\text{con}} + \alpha \mathcal{L}_{\text{cen}} + \beta \mathcal{L}_{\text{qua}} \tag{15}$$

in which α is used to balance the contrastive loss and center contrastive loss and β is the quantization penalty parameter.

C. Optimization

After defining the loss function, combined with (1), (2), (4), (5), and (15), the total loss function can be written as

$$\begin{aligned} \min_{\mathbf{W},\mathbf{v}} \mathcal{L} &= \mathcal{L}_{\text{con}} + \alpha \mathcal{L}_{\text{cen}} + \beta \mathcal{L}_{\text{qua}} \\ &= \sum_{i\in Q} \frac{-1}{n(S(i)) + \varepsilon} \sum_{s\in S(i)} \log \frac{\exp\left(\mathbf{z}_i \cdot \mathbf{z}_s^T/\tau\right)}{\sum_{r\in R(i)} \exp\left(\mathbf{z}_i \cdot \mathbf{z}_r^T/\tau\right)} \\ &\quad + \alpha \sum_{i=1}^{N} \sum_{c=1}^{C} l(\mathbf{z}_i, \mathbf{p}_c) + \beta \frac{\sum_{i=1}^{N} \sum_{k=1}^{K} (|h_{ik}| - 1)^2}{N} \end{aligned} \tag{16}$$

where $\mathbf{W} = \{\mathbf{W}_g, \mathbf{W}_l, \mathbf{W}_h\}$ and $\mathbf{v} = \mathbf{v}_h$. The backpropagation algorithm is selected to optimize network parameters $\mathbf{W}, \mathbf{v}$ in an end-to-end manner. The equation for gradient descent can be expressed as

$$\mathbf{W}_g \leftarrow \mathbf{W}_g - \gamma \nabla_{\mathbf{W}_g}(\mathcal{L}) \tag{17}$$

$$\mathbf{W}_l \leftarrow \mathbf{W}_l - \gamma \nabla_{\mathbf{W}_l}(\mathcal{L}) \tag{18}$$

$$\mathbf{W}_h \leftarrow \mathbf{W}_h - \gamma \nabla_{\mathbf{W}_h}(\mathcal{L}) \tag{19}$$

$$\mathbf{v}_h \leftarrow \mathbf{v}_h - \gamma \nabla_{\mathbf{v}_h}(\mathcal{L}) \tag{20}$$

where γ is the learning rate.

IV. Experiments

In this section, experiments are carried out to verify the effectiveness of the proposed framework. In Section IV-A, three benchmark datasets are introduced for conducting the experiments. In Section IV-B, experimental settings are described. Section IV-C shows the experimental results. And several important hyperparameters of the loss function are analyzed in Section IV-D.

A. Data Introduction

Three commonly used remote sensing retrieval datasets are used to verify the performance of retrieval methods, and their specific introductions are as follows.

The University of California, Merced dataset (UCMD) [39] was manually extracted from the USGS National Map Urban Area Imagery collection. It has 2100 scene images, which are divided into 21 land-use scene classes. Each class includes 100 images with a pixel size of 256 × 256 and a spatial

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Fig. 2. Examples of UCMD dataset.



Fig. 3. Examples of AID dataset.

resolution of 0.3 m. Some samples of this dataset are shown in Fig. 2.

The Aerial Image dataset (AID) [40] is a large-scale dataset for aerial scene classification. The image pixel size of AID is 600 × 600, and it contains 30 classes of scene images. The number of images in each class varies from 220 to 420 with a spatial resolution of 8–0.5 m. Some samples of this dataset are shown in Fig. 3.

NWPU-RESISC45 [41] is a publicly available benchmark for RSI scene classification. NWPU-RESISC45 contains 31 500 images with a size of 256 × 256, covering 45 scene classes, each with 700 images. The spatial resolution of the image varies from 30 to 0.2 m/pixel. Some samples of this dataset are shown in Fig. 4.

B. Experimental Settings

First, as shown in Fig. 1, the backbone network of our framework is AlexNet. The weight parameters of the first seven layers of AlexNet are initialized from the model which is pretrained by the ImageNet dataset and needs to be fine-tuned with the target dataset. The fc8 layer of AlexNet is replaced by a fully connected layer. Except for the first seven layers of AlexNet, other network structures are randomly initialized and need to be trained from scratch. Second, 70% of each class of image are randomly selected to construct the training set for each dataset, and the remaining 30% to construct the query set. It's worth noting that all images are resized to 224 × 224 pixels before being sent to the network. Third, the entire network is trained with a stochastic gradient descent (SGD) optimizer with a learning rate of 5×10^{-2}, momentum of 0.9, and weight decay of 5×10^{-2}. Mini-batch of images, temperature parameter τ, balancing parameter α, quantization penalty parameter β, and ε are set to 128, 0.1, 100, 0, and 10^{-5}, respectively. The setting analysis of these parameters is described in Section IV-D. All experiments are implemented under the PyTorch framework using Intel Xeon Gold 5218R

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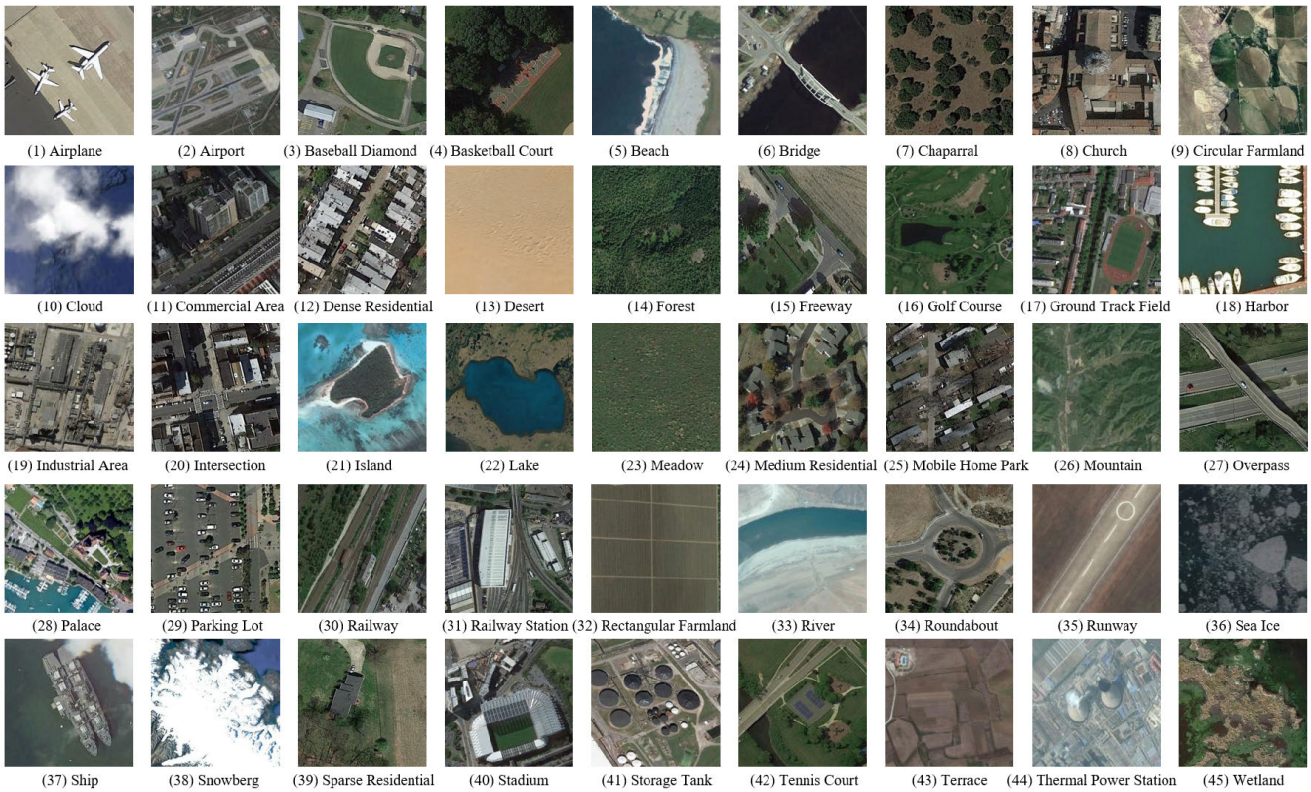


Fig. 4. Examples of NWPU dataset.

CPU and NVIDIA GeForce RTX 3080 GPU. All codes are written in Python 3.8.

Seven state-of-the-art hash retrieval methods are used here, including two remote sensing retrieval methods feature and hash (FAH) [12] and asymmetric hash code learning (AHCL) [16] and five natural image retrieval methods CSQ [37], DCH [36], DHN [35], deep pairwise-supervised hashing (DPSH) [42], and GreedyHash [43], to verify the effectiveness of our proposed method. Mean average precision (MAP), Precision@k, Recall@k, and precision and recall are the evaluation criteria to compare retrieval performance. They are calculated as follows.

1) *MAP:* Before acquiring the MAP, the Hamming distance between the query images and the database images needs to be calculated, and sorted in ascending order, which helps get the average precision (AP). AP can be expressed as

$$\mathrm{AP} = \frac{1}{E}\sum_{e=1}^{V}\mathrm{Precision}(e)\sigma(e) \tag{21}$$

where E is the number of images in the database that have the same label as the query image and V denotes the volume of the database. Precision(e) represents the precision of the top-e images of the sorted list. $\sigma(e) = 1$ denotes the eth retrieved image has the same label as the query image, and otherwise $\sigma(e) = 0$. Then, MAP can be obtained by calculating the mean of the AP from all the query images

$$\mathrm{MAP} = \frac{1}{Q}\sum_{i=1}^{Q}\mathrm{AP}_i \tag{22}$$

where Q is the number of query images.

2) *Precision@k:* This metric measures the precision value of the top-k retrieved images, which is defined as

$$\mathrm{Precision@}k = \frac{a}{k} \tag{23}$$

where k and a are the numbers of all images and similar images to the query image in the top-k list, respectively.

3) *Recall@k:* It computes the recall rate between the number of similar images to the query image in the top-k retrieved image and all similar images in the database, which is defined as

$$\mathrm{Recall@}k = \frac{a}{e} \tag{24}$$

where e and a are the numbers of similar images in the database and the top-k retrieved images, respectively.

4) *Precision and Recall:* The precision–recall curve is a popular measure of retrieval performance. The size of the area enclosed by it and the horizontal and vertical axes determines the retrieval performance.

C. Computational Complexity

The computational complexity of SCFR can be divided into two main parts. The first part is dependent on the SCFR network, which includes the global module, local module, and hash learning module. The second part of the computational complexity of SCFR pertains to the optimization of the objective function. The computational complexity of the comparison loss function is $\mathcal{O}(\mathcal{L}_{\mathrm{con}}) = N^2K$, while that of the center contrastive loss cost is $\mathcal{O}(\mathcal{L}_{\mathrm{cen}}) = NCK$. The computational complexity of quantization loss is $\mathcal{O}(\mathcal{L}_{\mathrm{quna}}) = NK$. Therefore, the computational complexity of the objective function can be expressed as $\mathcal{O}(\mathcal{L}) = NK(N + C + 1)$.

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TABLE I
MAP VALUE UNDER DIFFERENT BITS ON DIFFERENT DATASETS

Methods	UCMD				AID				NWPU-RESISC45			
	32 bits	64 bits	128 bits	256 bits	32 bits	64 bits	128 bits	256 bits	32 bits	64 bits	128 bits	256 bits
Ours	**97.65**	**97.95**	**98.29**	**98.14**	**93.79**	**94.42**	**94.59**	**94.53**	**90.48**	**91.11**	**91.78**	**91.79**
FAH	93.73	95.87	96.44	97.06	92.00	91.99	92.71	93.24	89.48	89.83	90.77	90.68
AHCL	96.02	96.70	-	-	92.46	92.75	-	-	86.51	90.13	-	-
CSQ	97.64	97.84	97.76	97.62	92.99	93.42	93.34	93.11	89.28	89.60	89.33	89.37
DCH	97.27	97.13	96.20	95.79	91.91	91.71	91.27	91.14	88.78	87.99	87.63	87.17
GreedyHash	96.16	95.60	97.05	93.28	92.39	93.24	93.73	94.18	89.53	90.59	90.83	91.19
DPSH	94.02	96.92	97.27	97.21	92.09	92.25	92.45	92.24	67.50	88.85	88.01	85.62
DHN	93.97	96.99	96.55	97.03	90.90	92.53	92.55	92.16	67.80	84.06	83.72	81.67

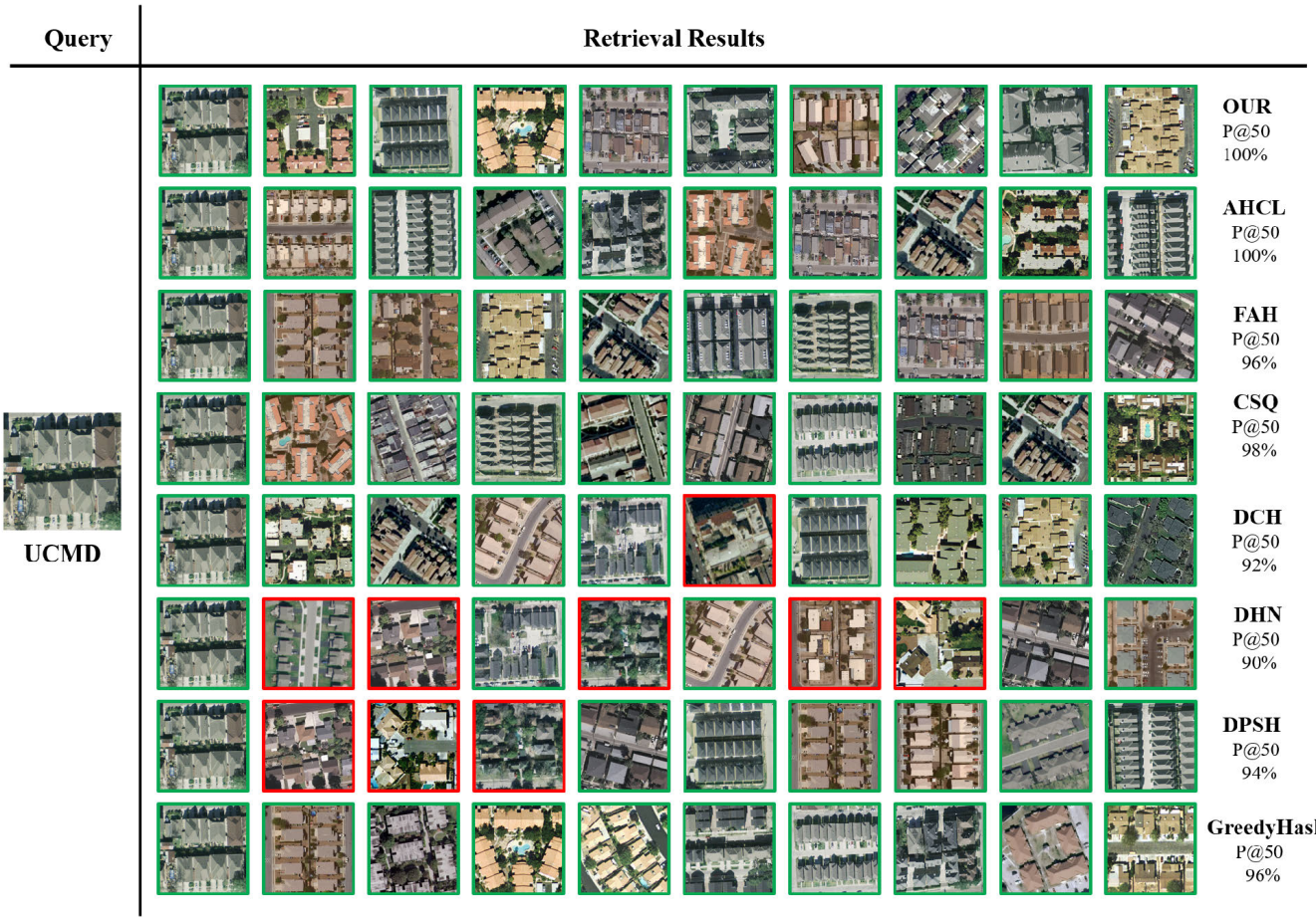


Fig. 5. Query example with the top-10 retrieved images and Precision@50 on the UCMD dataset.

D. Retrieval Performance

To verify the effectiveness of our proposed method, seven state-of-the-art hash retrieval methods are used for evaluating the retrieval performance of the hash codes. Figs. 5–7 display several query examples from the query set on different methods, where each row represents the retrieved result for one method, including top-10 retrieved images and the value of Precision@50. It is worth noting that the green box indicates a correct query and the red box indicates an incorrect query. From the figures, it can be seen that our method returns all correct top-10 retrieved images and the highest value of Precision@50 compared with other methods, which demonstrates the advantages of our method.

In Fig. 5, DHN and DPSH show poorer retrieval results compared with other methods, and the Precision@k values of FAH and CSQ are high and the Precision@k values of our method and AHCL reach 100% for UCMD dataset. In addition, Recall@k, Precision@k, and precision–recall curves for UCMD dataset are drawn in Fig. 8. It can be concluded that each method achieves excellent performance, among which our proposed method and CSQ [37] have more obvious advantages. In Fig. 6, we can observe that the retrieval performance of DHN, DPSH, and GreedyHash on the AID dataset is significantly reduced compared with the UCMD dataset. AHCL, FAH, and CSQ still obtain competitive retrieval results, but there are several false positives in their retrieval results compared with our method.

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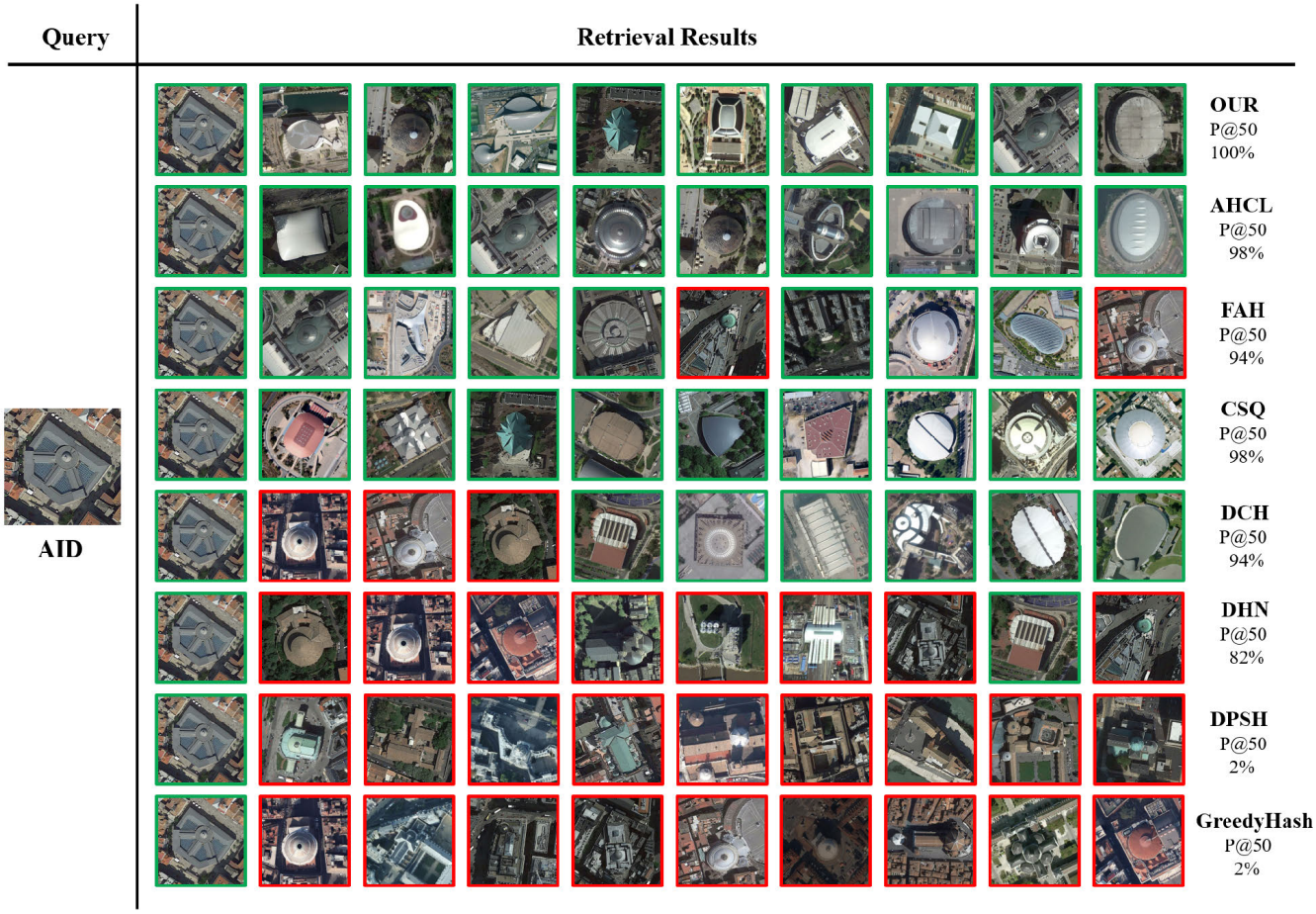


Fig. 6. Query example with the top-10 retrieved images and Precision@50 on the AID dataset.

Recall@k, Precision@k, and precision–recall curves for the AID dataset also show in Fig. 9. The curves illustrate that our method can get the highest precision values compared with other methods under the same recall values. In Fig. 7, we can see that FAH and DHN obtain the worse result results, while other methods perform well for the NWPU-RESISC45 dataset. And our method still performs the best. In addition, from Fig. 10, natural image retrieval methods based on hash learning have a prominent drop on three experimental curves compared with other datasets. On the contrary, RSIR methods based on hash learning maintain satisfactory retrieval results.

MAP is the primary quantitative metric for retrieval performance, which is shown in Table I. From the results, we can conclude as follows.

1) Under the same data, different bits of hash codes bring different results. In general, shorter hash code bits have worse experimental results due to greater loss of information.
2) The natural image retrieval method based on hash learning can have good retrieval performance when transferred to RSIs.
3) The proposed method obtains the highest MAP with different hash bits on three datasets.

Based on the above analyses, our proposed method achieves the best performance compared with other methods. Finally, we present the convergence curves of the loss function during the training process using 128-bit hash codes in Fig. 11. The general trend of the loss function convergence curve on all three datasets shows a decreasing trend, indicating that the model gradually converges during the training process. On the UCMD and AID datasets, the loss function curves tend to flatten, suggesting that the models eventually converge to a local optimum. However, during training on the NWPU-RESISC45 dataset, the convergence curve fluctuates up and down, indicating that the step size is too large and that the learning rate needs to be adjusted.

E. Parameter Analysis

In our loss function, there are three key hyperparameters α, Δ, and β that need to be tuned. α and β are used to balance contrastive loss, center contrastive loss, and quantization loss. And the parameter Δ defines how close a similar and dissimilar item can be. In the following, the influence of different values of α and Δ on the retrieval results is analyzed in detail.

Fig. 12(a) shows retrieval performance under different α on three datasets, where Δ is set to 0.5. The observation is that our method has similar experimental effects on the three datasets under different α. when the value of α changes from 0 to 1,

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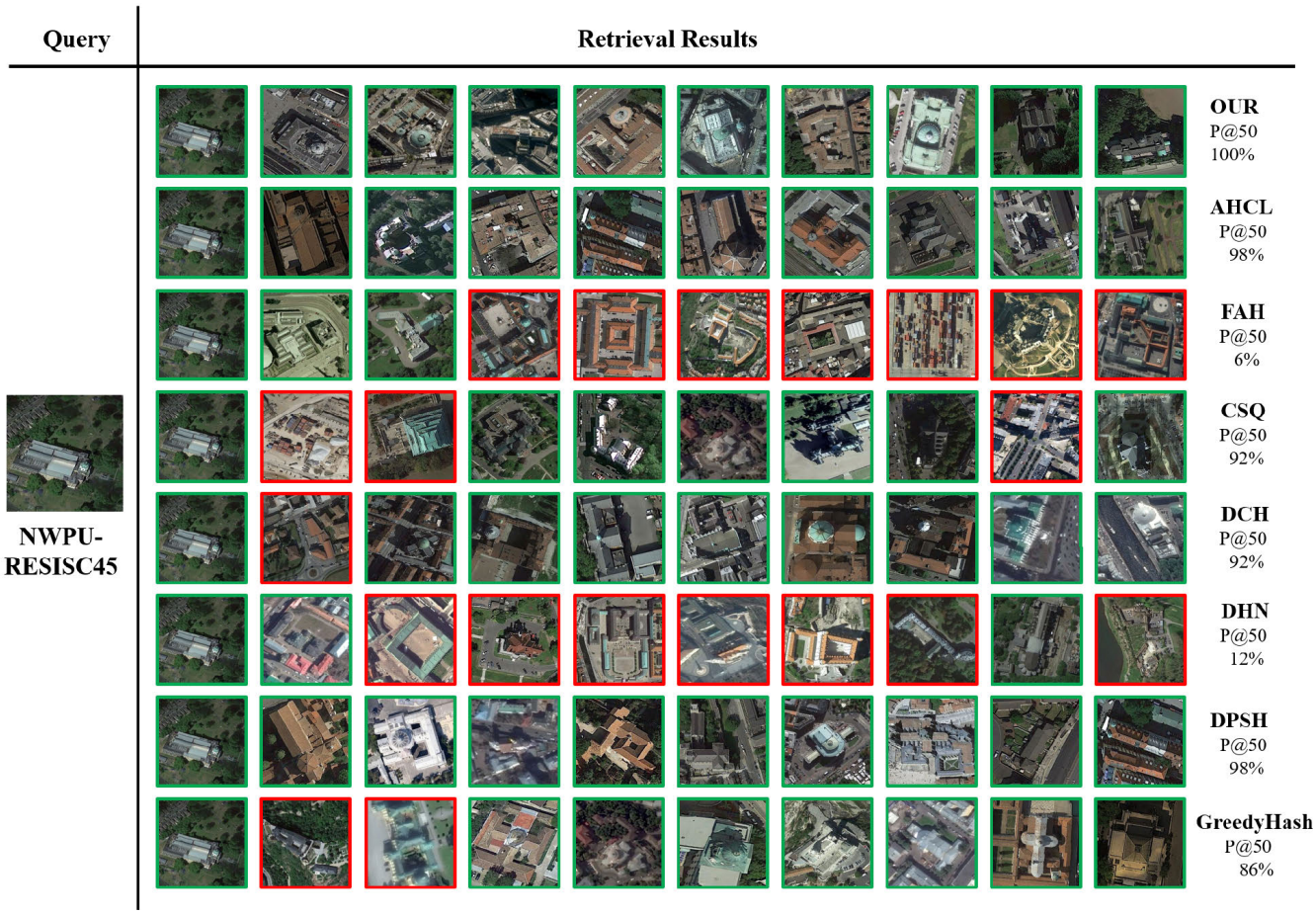
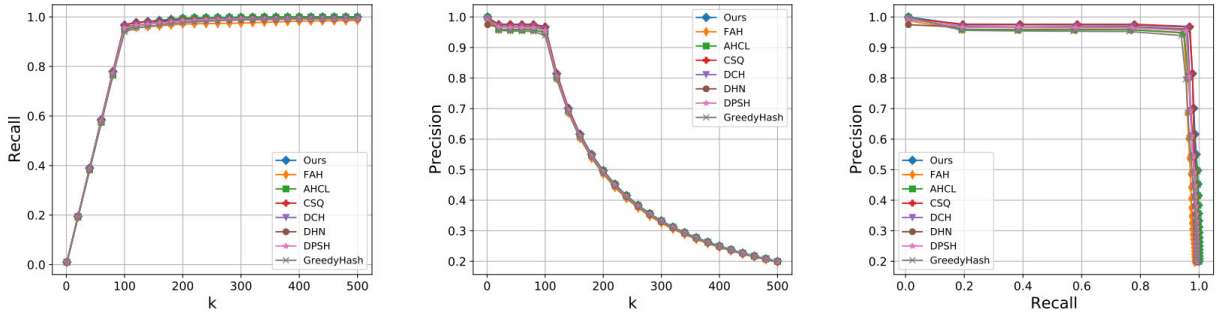
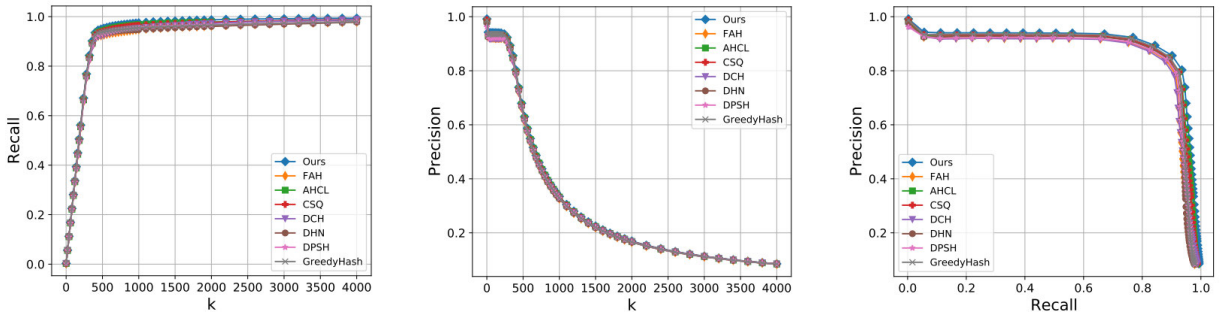


Fig. 7. Query example with the top-10 retrieved images and Precision@50 on the NWPU-RESISC45 dataset.



(a) (b) (c)

Fig. 8. Retrieval results on UCMD with 64-bit hash code. (a) Recall@k. (b) Precision@k. (c) Precision and recall.

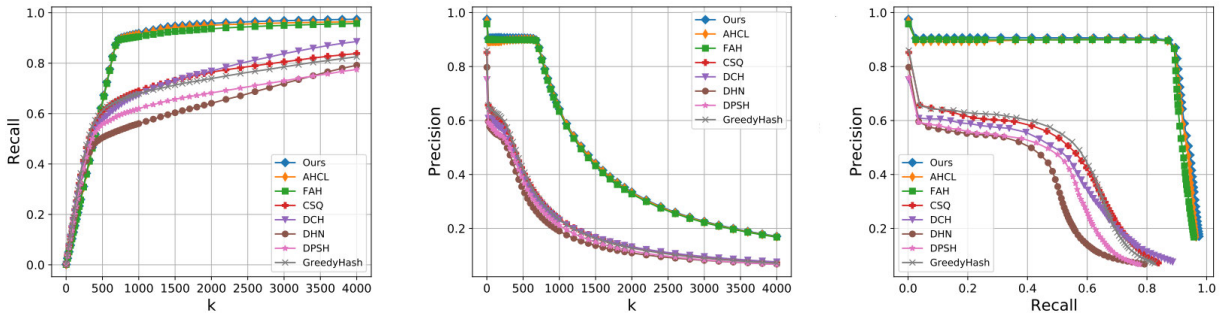


(a) (b) (c)

Fig. 9. Retrieval results on AID with 64-bit hash code. (a) Recall@k. (b) Precision@k. (c) Precision and recall.

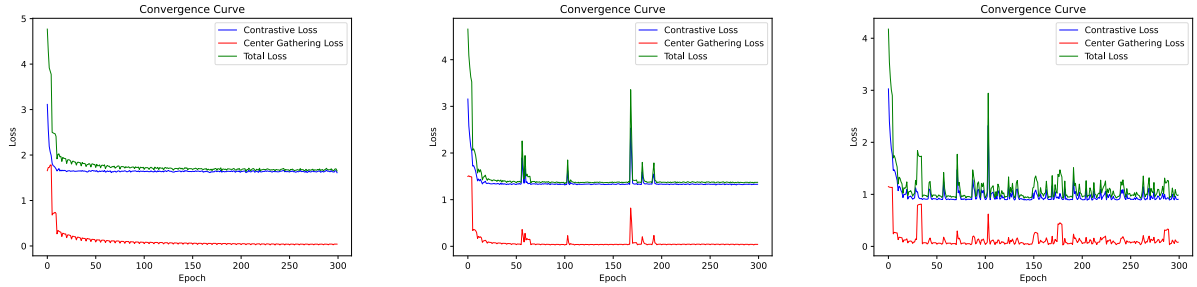
the MAP value is significantly improved on the three datasets. The main reason is that center contrastive loss effectively gathers features of the same class, which makes features more compact. As it can be seen from Fig. 12(a), with the increase

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(a) (b) (c)

Fig. 10. Retrieval results on NWPU-RESISC45 with 64-bit hash code. (a) Recall@k. (b) Precision@k. (c) Precision and recall.



(a) (b) (c)

Fig. 11. Convergence curve during training with 128-bit hash code. (a) UCMD. (b) AID. (c) NWPU-RESISC45.

TABLE II
MAP VALUE UNDER DIFFERENT METHODS ON DIFFERENT DATASETS

	UCMD				AID				NWPU-RESISC45			
	32 bits	64 bits	128 bits	256 bits	32 bits	64 bits	128 bits	256 bits	32 bits	64 bits	128 bits	256 bits
Method-1	97.60	96.75	96.76	96.63	93.17	93.85	93.79	93.44	89.78	90.23	90.10	89.37
Method-2	**97.67**	98.04	98.10	98.25	93.34	93.93	94.13	94.18	90.17	90.85	91.09	91.39
Method-3	97.63	**98.38**	**98.31**	**98.44**	**93.82**	94.07	94.59	94.53	90.39	91.09	91.57	**91.90**
Method-4	97.65	97.95	98.29	98.14	93.79	**94.42**	**94.59**	**94.53**	**90.48**	**91.11**	**91.78**	91.79
Method-5	97.20	97.68	97.76	98.27	93.67	94.01	94.34	94.46	89.89	91.02	91.35	91.56

of α, the MAP values can reach the peak value at $\alpha = 100$. Thus, α is set to 100 as the optimal value for the three datasets. Fig. 12(b) shows retrieval performance under different β on three datasets. What is clear is that as β increases from 0, the retrieval performance does not improve or even decrease. Therefore, we set from β to 0. In addition, Fig. 12(c) shows the effects of Δ on MAP for the three datasets, where α is set to 0.5. From this figure, it can be seen that the MAP reaches the maximum value when Δ is 0.5 for all datasets.

F. Ablation Study

In this section, we explore the influence of each component of our proposed method on the experimental results. There are contrastive loss, center contrastive loss, quantization loss, and fusion modules of global and local features. Therefore, five different methods are established for ablation study, which are as follows.

1) *Method-1:* AlexNet + classification loss.
2) *Method-2:* AlexNet + contrastive loss.
3) *Method-3:* AlexNet + contrastive loss + center contrastive loss.
4) *Method-4:* AlexNet + contrastive loss + center contrastive loss + fusion module.
5) *Method-5:* AlexNet + contrastive loss + center contrastive loss + fusion module + quantization loss.

Method-1 optimizes AlexNet with classification loss as the baseline. First, Table II shows that each component of our proposed method positively contributes to the retrieval performance except for the quantization loss which is mostly used in CBIR-based deep hashing learning. Hence, quantization loss may be unnecessary in CBIR based on deep hashing learning. Second, Method-2 has a higher MAP value than the baseline Method-1, which means that contrastive loss outperforms classification loss in CBIR tasks. Contrastive loss constraint ensures that the features of the same label are close, and the features of different labels are far away, resulting in similar properties of the features after binarization. Third, comparing the results of Method-2 and Method-3, center contrastive loss brings obvious improvement to the experimental results. This is because the loss of central constrains the features with the same semantic content, making them close to their corresponding feature centers and making the feature set more compact. Method-4 adds a feature fusion module, which improves the MAP value in some cases. It denotes that the fusion of global features and local features enhances the representation ability of features. In Method-5, the quantization loss function used

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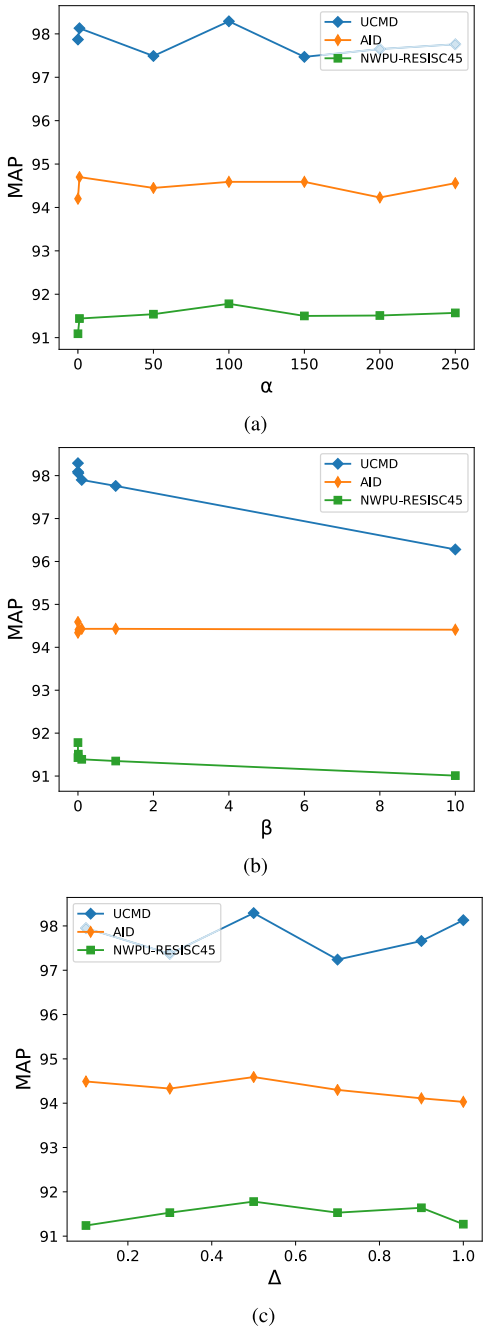


Fig. 12. Retrieval performance under different parameters on three datasets. (a) α. (b) β. (c) Δ.

in most hash retrieval is added to our constraints. And what you can see is the falling of the MAP value, which proves that the hash quantization function may not be necessary for the hash retrieval method. This is consistent with the result of our analysis in Fig. 12(b).

V. Conclusion

At present, most RSIR methods based on deep hashing only take global features as the final descriptor, which lacks the ability to discriminate images with the same semantic information but different visual representations. Therefore, a fusion module is proposed to combine local features and global features in this article. To achieve the feature distribution of intraclass compactness and interclass separability, supervised contrastive learning is introduced to optimize deep neural networks. In addition, we calculate the feature center of each class in the process of iterative network optimization and then use the feature centers to optimize the center contrastive loss, which can constrain the features of the same class more compactly. Extensive experimental results validate the effectiveness of our solution and show that our model achieves state-of-the-art image retrieval performances on three RSI datasets.

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